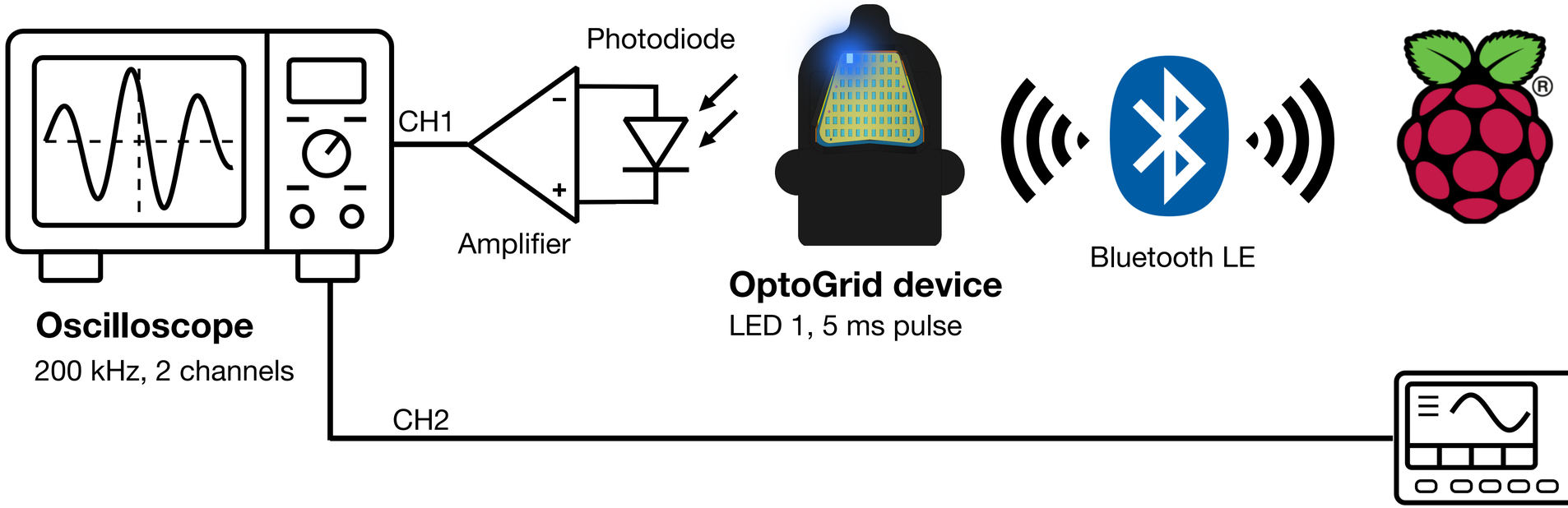
Trigger-to-light latency measurement



Test sequence

1. Function generator sends a 1 ms trigger pulse to the Raspberry Pi and oscilloscope CH2.
2. Raspberry Pi detects the pulse on GPIO 21 and sends a Bluetooth LE trigger command.
3. LED 1 emits a 5 ms blue light pulse, detected by a photodiode on oscilloscope CH1.
4. Latency = first light (CH1) – trigger onset (CH2), over ~1,200 trials.

- *Kunxin (Jessica) Wu, Sainsbury Wellcome Centre,12/10/2026*